SMART RECYCLING ROBOT

FOR AUTOMATED WASTE SORTING

Automating Smarter Tasks

The Smart Recycling Robot project is developing an autonomous robot using color and distance sensors to address the common issue of improper waste separation and resulting contamination in recycling. This robot will use intelligent decision-making and simple robotics functions like movement and obstacle avoidance to identify and sort different recyclable materials such as plastic, paper, or metal into appropriate bins based on color detection and programmed logic, thus demonstrating a practical, real-world application of intelligent robotics to improve waste management efficiency and support environmental sustainability.

Problem Statements

Current recycling efforts are hindered by widespread contamination from incorrect waste disposal, inefficient and labor-intensive manual sorting processes, and a significant gap in affordable, automated technology for accurate waste segregation.

Objective

This project aims to design and build an autonomous robot that uses sensors to detect common recyclable materials, makes intelligent sorting decisions, navigates its environment to avoid obstacles, and automatically deposits waste into the correct bins, thereby promoting recycling efficiency and environmental awareness.

MODULES

- Material Detection Module: We are using the color sensor to identify different recyclable materials such as paper, plastic, and metal.

- Obstacle Avoidance Module: Utilizes the ultrasonic sensor to detect and avoid obstacles while moving.

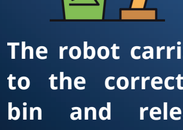
- Sorting Module: System will automatically sorts and deposits each material into the correct bin based on detection results.

- Movement Control Module: Controls the robot's forward, backward, and turning motions using motor commands.

- Programming Module: Integrates all functions through programming in Lego Mindstorm EV3 Education (or an equivalent platform) to enable autonomous operation.

How It's Work?

Line following



The robot follows a marked path using a light sensor to guide its movement along a black line.

Black Line Detect

The color sensor detects when the robot is correctly aligned with the line and adjusts direction if needed.

Detected Object

When an object (e.g., plastic, paper, or metal) is placed in front, the robot stops and uses the color sensor to identify the material type.

Grab the Object

The robot uses its gripper or claw mechanism to securely pick up the detected object.

Sort into Bin

The robot carries the object to the correct color-coded bin and releases it for proper recycling.

Reverse & Resume

After sorting, the robot reverses, returns to the line, and resumes following the path to await the next object.

METHODOLOGY

1. Hardware Assembly
   - Assemble the robot chassis using LEGO Mindstorms Education EV3 Core Set.
   - Mount the EV3 Light Sensor to detect material differences using reflected light intensity.
   - Attach the EV3 Ultrasonic Sensor for obstacle detection and bin distance measurement.
   - Configure motors for robot movement and sorting mechanism.

2. Programming Implementation
   - Use LEGO Mindstorms EV3 Education software (Scratch-like block programming).
   - Develop movement control blocks for forward, turn, and stop actions.
   - Program ultrasonic-based obstacle avoidance to prevent collisions.
   - Program light sensor detection to classify materials based on brightness levels.

3. Integration and Testing
   - Combine all sensor readings into a single control program.
   - Test robot movement, object detection, and sorting individually.
   - Perform system testing with different materials and adjust thresholds for accuracy.

Conclusion

This project showcases how robotics can enhance recycling efficiency by combining sensor-based detection and intelligent sorting. It provides a scalable solution for waste management that can be adapted for schools, public facilities, and recycling centers, setting a benchmark for future applications in environmental sustainability. Additionally, this technology can greatly benefit education, especially in promoting environmental awareness among students, making recycling more engaging and easier to practice daily.

Workshop 2 Group 7

SUPERVISOR NAME: PROF DR. BURHANUDDIN MOHD. ABOOBAIDER

GROUP MEMBER: ALICIA - B032420018, MUHAMMAD AMAR SYAKIR - B032310679, IRDINA HADIFAH - B032310829, RIVAN RAJ - B032410338

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FAKULTI KECERDASAN BUATAN DAN KESELAMATAN SIBER
UNIVERSITI TEKNIKAL MALAYSIA MELAKA